

Bayesian Prediction and Model Comparison in a Schizophrenia Treatment Study

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Topic

This project will use Bayesian data analysis to study longitudinal symptom trajectories in a randomized clinical trial of an antipsychotic drug versus placebo for patients with schizophrenia. The focus is on modeling how mean symptom severity changes over time under each treatment, accounting for between-patient variability, and using the posterior distribution to make predictive and comparative statements about treatment effects.

Research Questions

1. Were the drug and placebo groups comparable in mean baseline symptom severity?
2. How does mean symptom severity evolve over time in each group, and what is the posterior distribution of the difference in improvement between drug and placebo when we allow for patient-specific variability?
3. For a new patient starting treatment, what is the posterior predictive distribution of symptom severity at Week 6 under drug vs. placebo, and what is the probability of achieving a clinically meaningful level of improvement under each treatment?
4. For change from baseline (Week 0) to Week 6, which of several normal regression models (e.g., intercept only, +treatment, +treatment and baseline severity) is best supported by the data according to Bayesian model comparison?

Dataset

The data come from a schizophrenia clinical trial, referenced in the last section. Each row corresponds to a patient's visit at a particular week.

Key variables of use are:

- **id**: patient identifier (used for repeated measures / random effects)
- **Week**: time in weeks since baseline 0, 1, 3, 6, *etc*
- **TxDrug**: treatment indicator (drug vs. placebo)
- **imps79**: 1–7 symptom severity scale where higher values indicate more severe illness; continuous symptom severity score (higher = worse symptoms)
- **SqrtWeek** = $\sqrt{\text{Week}}$ to allow mild non-linearity in time

- Change scores (e.g., baseline to Week 1, baseline to Week 6) for some potential secondary analyses

I will restrict attention to patients with observed baseline and at least one follow-up IMPS79 measure. I will treat IMPS79 as approximately continuous to fit within the normal-model framework used in the course.

Models/distributions

All models will assume normal errors with conjugate or semi-conjugate priors based on the guidance from the lecture notes and to keep things simple.

Planned models:

1. Baseline comparison model:

This section tends to the 1st research question.

- Data: baseline (Week 0) IMPS79 only
- Model: two-group normal model for mean baseline IMPS79 in drug vs placebo, with normal priors on group means and inverse-gamma priors on variances.
- Target: posterior for the difference in baseline means

2. Hierarchical regression with random intercepts:

This section tends to the 2nd research question.

- Outcome: IMPS79.
- Fixed effects: Week, TxDrug, Week $\times$ TxDrug (and possibly SqrtWeek)
- Random effects: patient-specific intercepts $b_i \sim N(0, \tau^2)$
- Errors: $y_{ij} \mid \beta, b_i, \sigma^2 \sim N(x_{ij}^\top \beta + b_i, \sigma^2)$
- Priors: normal priors for β , inverse-gamma priors for σ^2 and τ^2
- Fitting: Gibbs sampler using full conditionals derived from the hierarchical normal model

3. Posterior prediction for new patients:

This section tends to the 3rd research question.

- Using draws from the posterior of the hierarchical model, I will simulate predictive IMPS79 values at Week 6 under drug vs placebo for new patients.
- I will summarize:
 - Posterior predictive means and credible intervals.
 - $P(\text{IMPS79 at Week 6} \leq c \mid \text{Drug})$ vs. placebo for a chosen threshold c . Preferably, $c = 3$ for mildly ill or better.

4. Model comparison:

This section tends to the 4th research question.

- Define $\Delta_i = \text{IMPS79}_{i,6} - \text{IMPS79}_{i,0}$ for patients with both visits.
- Compare three normal regression models for Δ_i :
 - Model A: $\Delta_i = \alpha_0 + \varepsilon_i$ (no treatment effect).
 - Model B: $\Delta_i = \alpha_0 + \alpha_1 \text{TxDrug}_i + \varepsilon_i$.
 - Model C: $\Delta_i = \alpha_0 + \alpha_1 \text{TxDrug}_i + \alpha_2 \text{Baseline}_i + \varepsilon_i$
- Use the closed-form marginal likelihood for the normal regression model (from the regression notes) to compute posterior model probabilities $p(M_k | y)$ and identify which specification is best supported.

All parameter inference will be via Monte Carlo (Gibbs sampler) with convergence diagnostics (trace plots, autocorrelation, etc.)

What questions and/or concerns do you have about the project?

- Normality of IMPS79: IMPS79 is an ordinal-like scale; treating it as continuous with normal errors is a simplifying assumption. I will check residuals and discuss this limitation, but I am intentionally staying within the normal-likelihood framework of the course.
- Missing data / irregular visit times: Some patients may have missing visits. I plan to use all available observed IMPS79 scores under a missing-at-random assumption; I may need guidance on whether this is acceptable or if simple sensitivity checks are expected.
- Computational details of the Gibbs sampler: For the hierarchical regression with random intercepts (and possibly `SqrtWeek`), I will derive full conditional distributions following the hierarchical and regression lecture notes. I may ask for feedback to confirm that my implementation and tuning (number of iterations, burn-in) are adequate.
- Scope vs. length: Depending on time, I may need to prioritize depth on fewer models (e.g., focus on one hierarchical specification (random intercepts for patients) plus prediction and a simple model comparison) rather than attempting too many variants.

Overall, I believe this project will let me apply the main ideas from the course—Bayesian normal models, hierarchical modeling, regression, Monte Carlo methods, and model comparison—to a scientifically meaningful question about treatment effects in a real clinical trial.

References

- Dataset: <https://hbiostat.org/data/repo/schizophrenia>